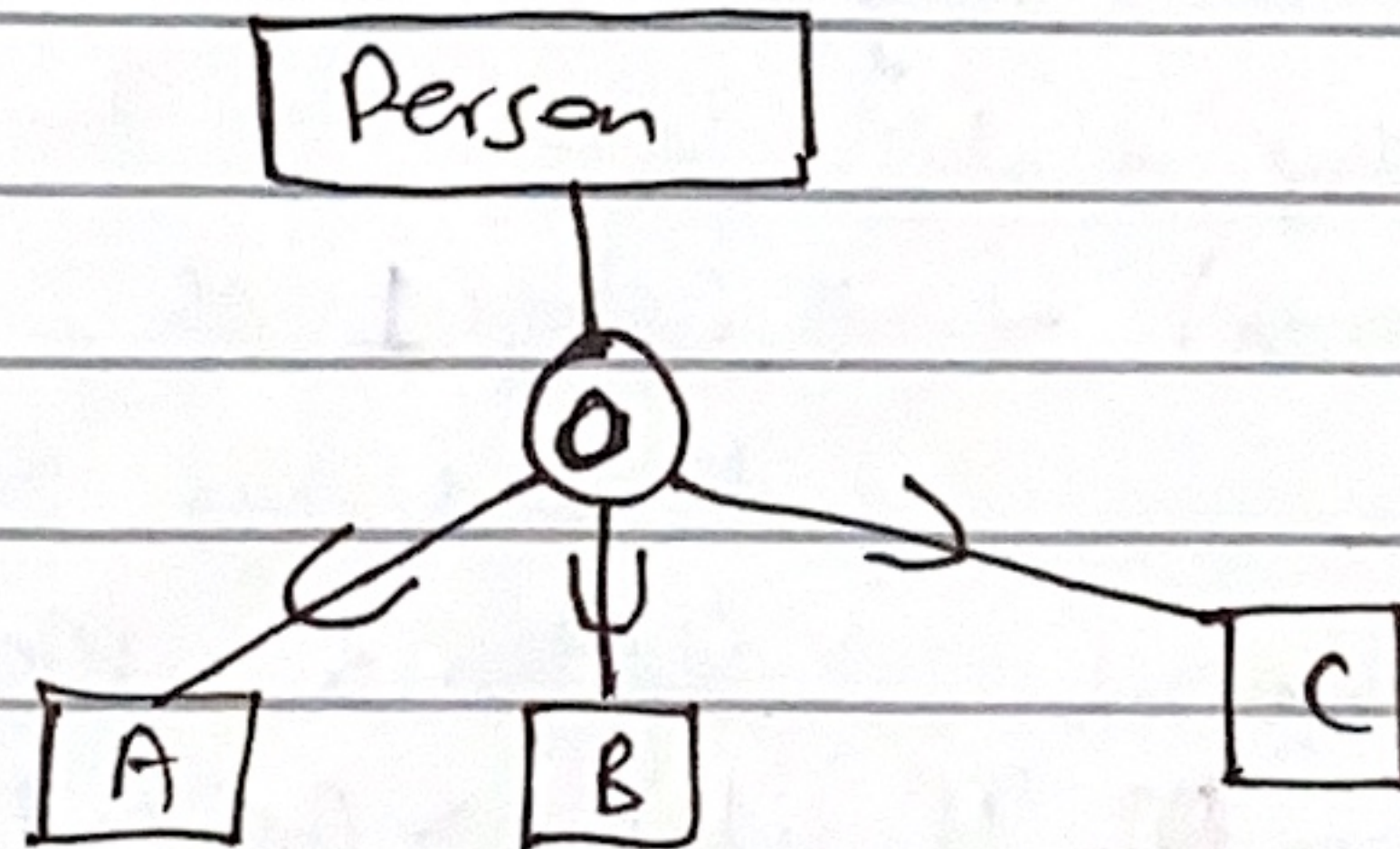


LECTURE-23

Mapping Inheritance Relations:



- ★ we'll judge which structure of table on basis of :
- Partial & Total dependency
 - Disjoint & overlapping.

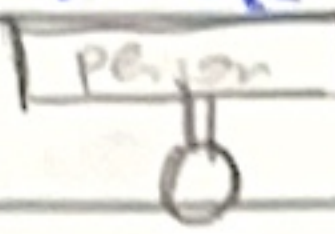
□ Which Tables are made in Super-Sub Class Cases.

① Make tables for all entities i.e.: Person, A, B, C.

The key of Person will be given to A, B, C with their unique elements. (Works for all cases)

- It works for all: overlapping, disjoint, partial/total dependency
- Needs a lot of JOIN for example display anything of A, B.

② We can just make tables of A, B, C and include all the attributes of Person in each. Ignore Parent class

- This would only work if there is total dependency b/w Person & A, B, C  means only A, B, C could be person and there couldn't be any other entity like D, E, F...
- If partial dependencies (single line) then it would fail in some cases
- It also only works for DISJOINT. Won't work for Overlapping.
- Advantage: Very less JOINS.

③ Make a single table of everything in one table which includes attributes of Person, A, B and C in single table along with an extra attribute PersonType.

- Disadvantage: A lot of NULLS.
- This won't work for overlapping as if we're an employee being a "manager" & "Engineer" then in JOBType; atomicity would be violated as
- It works for disjoint. And both partial & Total dependency

↓ solution of ③

④ Make a single table with everything in it and to allow overlapping and make Boolean attributes for JobTypes.

E.g: Technician (Yes/No), Engineer (Yes/No) etc.

- Won't work sustainably for partial dependency. (A work around would be to alter schema whenever a new job type appears).

NOTE: Whenever we've multiple inheritance (4th, 3rd gen. etc). we'll start making tables from bottom most relation inheritance.

- Also start looking from cases starting from ④, ③, ②, ①. to apply on tables

⇒ Eg of Multiple Inheritance on page # 37/45 & 38/45

- The "employee → staff, faculty and Student assistant" is converted into tables using Rule 3 (making single table).

- There are "Ra-flag" and "Ta-flag" ^{that it's ③ case} in Employee table since these two flags are from the bottom most inheritance of: "Student Assistant → Research-As and Teaching-As." where there is overlapping.